C1 Introduction

1.1 What is Pattern Recognition (PR)?

1.1.1 What is Pattern?

A pattern is **the opposite of a chaos**; it is an entity vaguely defined, that could be given a name."

There are various kinds of patterns:

1. **Visual patterns (视觉模式)** such as eyes, nose, mouth, face, fingerprint, etc.
2. **Temporal patterns (时序模式)** such as speech, audios, videos, data streams, etc.
3. **Logical patterns (逻辑模式)** such as characters, strings, images, etc.

1.1.2 What is Recognition?

identification of a pattern as a member of a category we already know, or we are familiar with.

Two types of recognition:

1. Classification (分类): Categories are **known** and the task is to assign a proper class label for each pattern.
2. Clustering (聚类): Categories are **unknown** and the task is to learn categories and group the patterns accordingly.

1.1.3 What is Pattern Recognition?

Pattern recognition is the procedure of **processing and analyzing diverse information** (numerical, literal, logical) characterizing the objects or phenomenon, so as to **provide descriptions, identifications, classifications and interpretations** for them.

It is the study of how machines can:

- **Perceive:** Observe the environment (i.e. interact with the real-world)
- **Process:** Learn to distinguish patterns of interest from their background
- **Prediction:** make sound and reasonable decisions about the categories of the patterns

1.2 Why do we need Pattern Recognition?

Pattern recognition is needed in designing almost all automated and intelligent systems:

1. Character Recognition
2. Speech Recognition
3. Fingerprint Recognition
4. Signature Identification
5. Face Detection
6. Text Categorization

All in all, pattern recognition is important, useful, attractive, but rather challenging.

1.3 HoW to perform Pattern Recognition?

Step I: Preprocessing (预处理)

Goal: Preprocess the image captured by the camera, such that subsequent operations could be simplified without losing relevant information

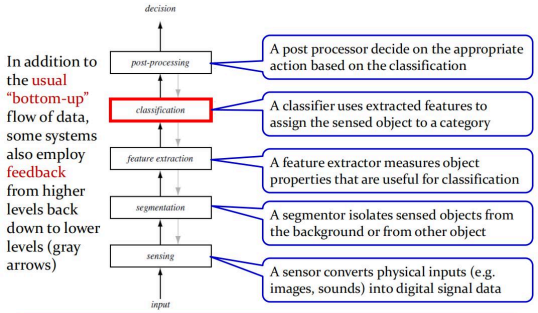
Step II: Feature Extraction (特征抽取)

Goal: Extract features (with good distinguishing ability) from the preprocessed image to be used for subsequent classification

Step III: Classification (分类)

Goal: To distinguish different types of objects (in this case, sea bass vs. salmon) based on the extracted features

input → segmentation → feature extraction → classifiation → post-processing → decision



1.3.1 如何实现好的泛化能力？

在 **Performance on the training set** 和 **Simplicity of the classifier** 之间做权衡

1.3.2 Design Cycle of PR System

The design of a PR system usually entails a number of different activities, such as **data collection, feature choice, model choice, classifier training, classifier evaluation.**

- **Data collection accounts for a large part of the cost of developing a PR system**
- **Feature choice and model choice are highly domain-dependent, where prior knowledge(先验知识) plays very important role**
- **Various activities may be repeated in order to obtain satisfactory results**

1.4 Important Issues in Pattern Recognition

1.4.1 Noise(噪声)

Any property of the sensed pattern which is not due to the true underlying model but instead to intrinsic randomness of the world or the sensors

1.4.2 Segmentation(分割)

Individual patterns have to be segmented for subsequent pattern recognition operations.

One of the deepest as well as hardest problems in pattern recognition:

- How can we segment the images without having categorized them firstly?
- On the other hand, how can we categorize the images without having segmented them firstly?

1.4.3 Data Collection(数据采集)

A sufficiently large and representative dataset needs to be collected.

1.4.4 Domain Knowledge(领域知识)

When training data is insufficient, prior knowledge is of paramount importance.

- Type I: Incorporate domain knowledge on the patterns themselves – Difficult!
- Type II: Incorporate domain knowledge on the pattern generation procedure.

1.4.5 Feature Extraction(特征抽取)

A domain-dependent problem which influences the classifier's performance

What kinds of features are promising?(什么样的特征是好特征)

Distinguishing Capability(具有区分能力): Whose values are very similar for objects in the same category, while very different for objects in different categories.

1.4.6 Pattern Representation(模式表示)

The most commonly used method is the statistical approach (feature vector). The ideal representation should ensure "small within-class variance and large between-class variance."

1.4.7 Missing Features(特征缺失)

In practical problems (such as occlusion), certain feature values may be missing.

1.4.8 Model Selection(模型选择)

- Each pattern recognition method employs certain model hypothesis
- Every pattern recognition problem has its own underlying true model

1.4.9 Overfitting(过配)

The model is overly complex, leading to perfect performance on the training set but poor generalization to new data (the test set).

1.4.10 Context(上下文)

- **Context:** Input-dependent information, other than from the pattern itself.
- The same pattern within different context might have different meanings

1.4.11 Classifier Ensemble(分类器集成)

Using multiple classifiers to collaboratively accomplish tasks to enhance generalization performance.

1.4.12 Costs and Risks(代价与风险)

In the real world, different erroneous decisions may have unequal costs. For example, in HIV diagnosis, the cost of misclassifying a "positive" as "negative" is far greater than misclassifying a "negative" as "positive".

1.4.13 Computational Complexity(计算复杂度)

How the complexity of the algorithm (in terms of time and storage) scales with the number of features, training samples, and classes. It is necessary to strike a balance between performance and computational simplicity.

Ch2 贝叶斯理论

2.1 分类中的贝叶斯公式（Bayesian formula）

$$P(\omega_j|x) = \frac{p(x|\omega_j)\cdot P(\omega_j)}{p(x)}$$

$P(\omega_j)$ ：先验概率 Prior Probability

$p(x|\omega_j)$ ：类别 j 的类条件概率密度 Class-conditional pdf，描述了在类别 j 中特征 x 的分布情况

$p(x) = \sum_{j=1}^{c} p(x|\omega_j)P(\omega_j)$ ：全概率公式，Total Probability Formula

2.2 贝叶斯决策

2.2.1 最小错误率决策 (Minimum-Error-Rate)

- 目标：做出决策以最小化分类错误的概率。
- 决策规则：于一个给定的特征x，计算它属于每个类别 ω_j 的后验概率 $P(\omega_j|x)$ (Posterior Probability) 。选择具有最大后验概率的那个类别。
- 决策：如果 $P(\omega_i|x) > P(\omega_j|x)$ 对于所有 $j \neq i$，则判别 x 属于 ω_i
- 如果我们决定 ω_i，那么犯错的概率就是 x 实际上属于任何其他类别的概率之和，

$$P(error|x) = \sum_{j\neq i} P(\omega_j|x) \qquad P(error|x) = 1 - P(\omega_i|x)$$

2.2.2 包含"代价 (Cost)"的贝叶斯决策 (一般情况)

条件风险的计算：

$$R(\alpha_i|x) = \sum_{j=1}^{c} \lambda(\alpha_i|\omega_j)\cdot P(\omega_j|x)$$

- 损失函数 (Loss Function) $\lambda(\alpha_i|\omega_j)$：当真实类别是 ω_j 时，我们采取了 α_i 动作所带来的代价或损失
- 条件风险 (Conditional Risk) $R(\alpha_i|x)$：在观察到 x 的情况下，采取动作 α_i 的期望损失
- **一般贝叶斯决策规则：**
  对于一个给定的 x，计算每一种动作 α_i 的条件风险 $R(\alpha_i|x)$，并选择那个风险最小的动作：$\alpha(x) = \arg\min_{\alpha_i\in\mathcal{A}} R(\alpha_i|x)$

特殊情况——二分类问题：

$$\boldsymbol{R(\alpha_1|x) = \lambda_{11}P(\omega_1|x) + \lambda_{12}P(\omega_2|x)} \qquad \boldsymbol{R(\alpha_2|x) = \lambda_{21}P(\omega_1|x) + \lambda_{22}P(\omega_2|x)}$$

似然比推导：

- 起始： $\boldsymbol{R(\alpha_1|x) < R(\alpha_2|x)}$
- 代入定义：$(\lambda_{21}-\lambda_{11})P(\omega_1|x) > (\lambda_{12}-\lambda_{22})P(\omega_2|x)$
- 应用贝叶斯定理 (Bayes theorem)：

$$(\lambda_{21}-\lambda_{11})\frac{p(x|\omega_1)P(\omega_1)}{p(x)} > (\lambda_{12}-\lambda_{22})\frac{p(x|\omega_2)P(\omega_2)}{p(x)}$$

$$(\lambda_{21}-\lambda_{11})p(x|\omega_1)P(\omega_1) > (\lambda_{12}-\lambda_{22})p(x|\omega_2)P(\omega_2)$$

- 整理成似然比形式：

$$\frac{p(x|\omega_1)}{p(x|\omega_2)} > \frac{\lambda_{12}-\lambda_{22}}{\lambda_{21}-\lambda_{11}}\cdot\frac{P(\omega_2)}{P(\omega_1)}$$

左边 $\frac{p(x|\omega_1)}{p(x|\omega_2)}$ 是似然比(Likelihood Ratio)，它只跟我们的类条件概率模型有关

右边是一个阈值，如果似然比大于这个阈值，我们就决策为 ω_1，否则决策为 ω_2

2.3 判别函数(Discriminant Function)：

- 决策规则：如果 $g_i(x) > g_j(x)$ 对于所有 $j \neq i$ ，则判别 x 属于 ω_i。
- $g_i(x) = \ln p(x|\omega_i) + \ln P(\omega_i)$

2.4 高斯密度 (Gaussian Density)：

2.4.1 单变量高斯 (Univariate)：

- 公式为： $p(x) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$

2.5 高斯密度下的判别函数：

情况一：$\Sigma_i = \sigma^2 I$ (协方差矩阵与类别无关，且特征不相关)

此时，判别函数为： $g_i(x) = w_i^t x + w_{i0}$

- 权值向量：$w_i = \frac{1}{\sigma^2}\mu_i$
- 偏置/阈值： $w_{i0} = -\frac{1}{2\sigma^2}\mu_i^t\mu_i + \ln P(\omega_i)$

情况二：$\Sigma_i = \Sigma$ (所有类别的协方差矩阵相同)

此时，判别函数为： $g_i(x) = w_i^t x + w_{i0}$

- 权值向量 $w_i = \Sigma^{-1}\mu_i$
- 偏置/阈值 $w_{i0} = -\frac{1}{2}\mu_i^t\Sigma^{-1}\mu_i + \ln P(\omega_i)$

情况三：Σ_i = 任意 (Arbitrary)(协方差矩阵对每个类别都不同)

此时，判别函数为： $g_i(x) = x^t W_i x + w_i^t x + w_{i0}$

- 二次矩阵：$\boldsymbol{W_i = -\frac{1}{2}\Sigma_i^{-1}}$
- 权值向量： $\boldsymbol{w_i = \Sigma_i^{-1}\mu_i}$
- 偏置/阈值： $\boldsymbol{w_{i0} = -\frac{1}{2}\mu_i^t\Sigma_i^{-1}\mu_i - \frac{1}{2}\ln|\Sigma_i| + \ln P(\omega_i)}$

Ch3 Maximum-Likelihood and Bayesian Parameter Estimation

1. 找到似然函数： $p(\mathcal{D}|\theta) = \prod_{k=1}^{n} p(x_k|\theta)$
2. 找到对数似然 (Log-Likelihood)函数：

$$l(\theta) = \ln[p(\mathcal{D}|\theta)] = \ln\left[\prod_{k=1}^{n} p(x_k|\theta)\right] = \sum_{k=1}^{n}\ln p(x_k|\theta)$$

3. 使梯度为 0：

$$\nabla_\theta l = \nabla_\theta\left[\sum_{k=1}^{n}\ln p(x_k|\theta)\right] = \sum_{k=1}^{n}\nabla_\theta \ln p(x_k|\theta) = \mathbf{0}$$

Maximum-Likelihood (ML) estimation (极大似然估计)

View parameters as quantities whose values are **fixed but unknown**.

Estimate parameter values by **maximizing the likelihood**(probability) of observing the actual training examples.

3.1 ML 应用于高斯分布

3.1.1 情况 1：只估计均值 μ (Σ 已知)

$$\ln p(x_k|\mu) = -\frac{1}{2}\ln[(2\pi)^d|\Sigma|] - \frac{1}{2}(x_k-\mu)^t\Sigma^{-1}(x_k-\mu)$$

- 求梯度(Find the gradient) $\nabla_\mu \ln p(x_k|\mu) = \mathbf{0}$：

$$\nabla_\mu l = \sum_{k=1}^{n}\nabla_\mu \ln p(x_k|\mu) = \sum_{k=1}^{n}\Sigma^{-1}(x_k-\hat{\mu}) = \mathbf{0}$$

得出： $\hat{\mu} = \frac{1}{n}\sum_{k=1}^{n} x_k$

3.1.2 情况 2：μ和 σ^2 均未知 (一维高斯)

$$p(x_k \mid \theta) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \quad \left(\theta = \begin{bmatrix}\theta_1\\ \theta_2\end{bmatrix} = \begin{bmatrix}\mu\\ \sigma^2\end{bmatrix}\right)$$

$$\ln p(x_k|\theta) : -\frac{1}{2}\ln(2\pi\theta_2) - \frac{1}{2\theta_2}(x_k-\theta_1)^2 。$$

求梯度 $\nabla_\theta \ln p(x_k|\theta)$:

$$\frac{\partial}{\partial\theta_1}(\dots) = \frac{1}{\theta_2}(x_k-\theta_1)$$

$$\frac{\partial}{\partial\theta_2}(\dots) = -\frac{1}{2\theta_2} + \frac{(x_k-\theta_1)^2}{2\theta_2^2}$$

设置 $\nabla_\theta l = \mathbf{0}$, 得出:

$$\cdot \sum_{k=1}^{n}\frac{1}{\hat{\theta}_2}(x_k-\hat{\theta}_1) = 0$$

$$\cdot \sum_{k=1}^{n}\left(-\frac{1}{2\hat{\theta}_2} + \frac{(x_k-\hat{\theta}_1)^2}{2\hat{\theta}_2^2}\right) = 0$$

结果：

$$\hat{\theta}_1 = \hat{\mu} = \frac{1}{n}\sum x_k \qquad \hat{\theta}_2 = \frac{1}{n}\sum(x_k-\hat{\theta}_1)^2, \text{ 即 } \hat{\sigma}^2 = \frac{1}{n}\sum(x_k-\hat{\mu})^2 \text{(样本方差)} 。$$

3.1.3 情况 3：μ 和Σ 均未知 (多维高斯)

结果：

均值 $\hat{\mu}: \hat{\mu} = \frac{1}{n}\sum_{k=1}^{n} x_k$

协方差 $\hat{\Sigma}: \hat{\Sigma} = \frac{1}{n}\sum_{k=1}^{n}(x_k-\hat{\mu})(x_k-\hat{\mu})^t$ (样本协方差矩阵)

3.2 贝叶斯估计

View parameters as **random variables** having some known prior distribution

Observation of the actual training examples transforms parameters' **prior distribution into posterior distribution (via Bayes theorem)**

阶段 I： 根据θ的先验概率密度函数，求出θ的后验概率密度函数

$$\boldsymbol{p(\theta|\mathcal{D}) = \frac{p(\theta,\mathcal{D})}{p(\mathcal{D})} = \frac{p(\theta)p(\mathcal{D}|\theta)}{\int p(\theta,\mathcal{D})\,d\theta} = \frac{p(\theta)p(\mathcal{D}|\theta)}{\int p(\theta)p(\mathcal{D}|\theta)\,d\theta}}$$

其中 , $\boldsymbol{p(\mathcal{D}|\theta) = \prod_{k=1}^{n} p(x_k|\theta)}$。

–输入： 训练集 $\mathcal{D}$

–先验概率密度: $\boldsymbol{p(\theta)}$

–参数化形式（似然函数）: $\boldsymbol{p(x|\theta)}$

–输出： 后验概率密度 $\boldsymbol{p(\theta|\mathcal{D})}$

阶段 II：根据θ的后验概率密度函数，求出 x 的类条件概率密度函数

$$\boldsymbol{p(x|\mathcal{D}) = \int p(x|\theta)p(\theta|\mathcal{D})\,d\theta}$$

–输入： 后验概率密度函数 $\boldsymbol{p(\theta|\mathcal{D})}$

–参数化形式: $\boldsymbol{p(x|\theta)}$

–输出： 类条件概率密度函数 $\boldsymbol{p(x|\mathcal{D})}$

阶段 III：分类 (Classification)

当得到 $\boldsymbol{p(x|\mathcal{D})}$ 后，可计算后验类别概率：

$$\boldsymbol{P(\omega_j|x,\mathcal{D}^*) = \frac{P(\omega_j)\,p(x|\omega_j,\mathcal{D}_j)}{\sum_{j=1}^{c} P(\omega_j)\,p(x|\omega_j,\mathcal{D}_j)}}$$

3.3 高斯情况下的贝叶斯估计 (均值 μ 未知)

单变量情况

- 目标参数：$\theta = \{\mu\}$ (未知的均值)。
- 已知条件：
  - σ_i^2 (数据的方差) 是已知的。
  - $p(\mu_i) \sim N(\mu_{0,i}, {\sigma_{0,i}}^2)$
  - $\bar{x_i} = \frac{(x_1+x_2+\dots+x_n)\in\mathcal{D}_i}{n}$
- 独立同分布 (i.i.d.)：假设数据 $\mathcal{D} = \{x_1,\dots,x_n\}$是独立同分布的，似然 $p(\mathcal{D}|\mu)$ 可以写成连乘形式：

$$\boldsymbol{p(\mu|\mathcal{D}) \propto p(\mu)\cdot\prod_{k=1}^{n} p(x_k|\mu)}$$

结论：

$$\frac{1}{\sigma_{n,i}^2} = \frac{1}{\sigma_{0,i}^2} + \frac{n}{\sigma_i^2} \qquad \mu_{n,i} = \sigma_{n,i}^2\left(\frac{n\bar{x}_i}{\sigma_i^2} + \frac{\mu_{0,i}}{\sigma_{0,i}^2}\right) \qquad \boldsymbol{p(x|\boldsymbol{D}_i) \sim N(\mu_{n,i}, \sigma_i^2 + \sigma_{n,i}^2)}$$

Ch4 非参数技术

4.1 Parzen Windows

该方法固定区域体积 V_n ：

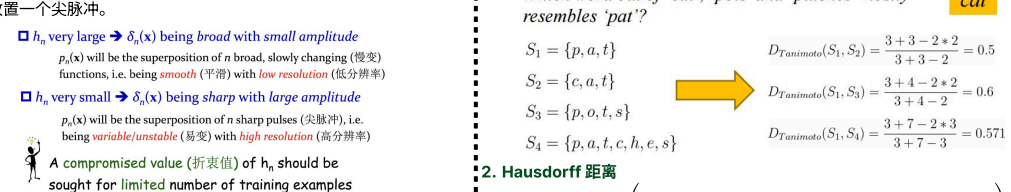
1. 定义区域： 假设$\mathcal{R}_n$ 是一个 d 维超立方体，边长为 h_n，因此体积 $V_n = h_n^d$
2. 计算 k_n： 我们使用一个窗口函数 $\varphi(\boldsymbol{u})$(也称为核函数) 来统计落入窗口的点。
   a. 一个简单的窗口函数是：当 x_i 落在以 x 为中心、体积为V_n的超立方体内时，$\varphi(\frac{x-x_i}{h_n}) = 1$ ，否则为 0。
   b. k_n 就是所有样本点的窗口函数值之和： $k_n = \sum_{i=1}^{n} \varphi(\frac{x-x_i}{h_n})$
3. 最终公式： 将 k_n 和 V_n 代入基本公式，得到 Parzen 密度估计：

$$p_n(x) = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{V_n}\varphi(\frac{x-x_i}{h_n})$$

a. 这本质上是 n 个插值函数（核）的叠加或平均。
b. 窗口函数 $\varphi(\cdot)$不必是立方体，它可以是任何概率密度函数 (pdf)，例如高斯函数。

4.1.1 窗口宽度 h_n 的影响

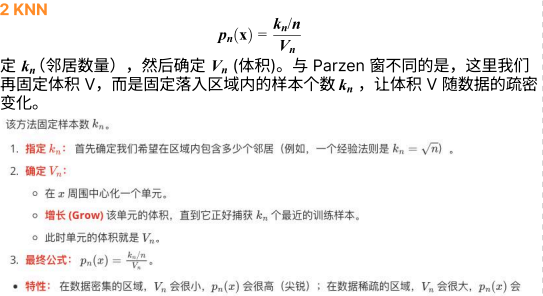
1. **情况一：** h_n 变大，那么窗口宽度就会**变大**，但总积分仍是 1，所以核函数高度(amplitude) 就会**变矮**。此时，$p_n(x)$ 估计非常平滑，但分辨率很低（慢变）。
2. **情况二：** h_n 变小，窗口宽度**变窄**，高度**变高，核函数 $\delta_n(x)$ 变得非常狭窄且尖锐**。此时，$p_n(x)$ 估计非常易变/不稳定（高分辨率），基本上是在每个数据点上放置一个尖脉冲。



4.2 KNN

$$p_n(x) = \frac{k_n/n}{V_n}$$

固定 k_n（邻居数量），然后确定 V_n (体积)。与 Parzen 窗不同的是，这里我们不再固定体积 V，而是固定落入区域内的样本个数 k_n ，让体积 V 随数据的疏密而变化。



4.3 最近邻 (NN) 分类规则

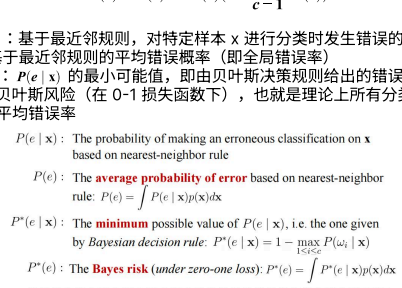
4.3.1 1-NN 规则（最近邻）

1. **给定：** n 个已标记的训练样本 $\mathcal{D}^n$ 。
2. **算法：**
   a. 对于一个新的测试样本 x，找到训练集中与 x 最近的那个样本 x' (例如使用欧氏距离)。
   b. 将 x' 关联的标签 θ' 赋给 x。
3. **Voronoi 维诺图：** 1-NN 规则将特征空间划分为 n 个单元（每个训练样本一个）。单元内的任何点都共享该样本的标签。
4. **错误率：** NN 规则的错误率 $P(e)$ 受贝叶斯风险 $P^*(e)$ 的限制：

$$P^*(e) \le P(e) \le P^*(e)(2 - \frac{c}{c-1}P^*(e))$$

其中：

- $P(e \mid \mathbf{x})$ ：基于最近邻规则，对特定样本 x 进行分类时发生错误的概率。
- $P(e)$：基于最近邻规则的平均错误概率（即全局错误率）
- $P^*(e \mid \mathbf{x})$： $P(e \mid \mathbf{x})$ 的最小可能值，即由贝叶斯决策规则给出的错误率。
- $P^*(e)$：贝叶斯风险（在 0-1 损失函数下），也就是理论上所有分类器能达到的最低平均错误率



4.3 距离度量 (Distance Metric)

4.3.1 距离属性

一个有效的距离度量 D(a, b) 必须满足：

1. 非负性： $\boldsymbol{D(a,b) \ge 0}$
2. 自反性： $\boldsymbol{D(a,b) = 0 \iff a = b}$
3. 对称性： $\boldsymbol{D(a,b) = D(b,a)}$
4. 三角不等式： $\boldsymbol{D(a,b) + D(b,c) \ge D(a,c)}$

4.3.2 常见的距离度量

闵可夫斯基距离 (L_k)： $L_k(a,b) = (\sum_{j=1}^{d} |a_j - b_j|^k)^{\frac{1}{k}}$

- k=1: 曼哈顿距离 (L_1) (Manhattan distance / city block)
- k=2: 欧氏距离 (L_2) (Euclidean distance)
- $k = \infty$: L_∞距离 ($max_j|a_j - b_j|$)

4.3.3 集合间的距离

1. Tanimoto 距离

$$D_{\text{Tanimoto}}(S_1, S_2) = \frac{n_1 + n_2 - 2n_{12}}{n_1 + n_2 - n_{12}}$$

- n_1：集合 S_1中的元素个数。
- n_2：集合 S_2中的元素个数。
- n_{12}：两个集合交集的个数（即两个集合共有的元素个数）。
- 结论：这个值越小，说明两个集合越相似（距离越近）；值越大，说明越不相似。

问题： 把单词看作字母的集合。在 'cat', 'pots', 'patches' 这三个词中，哪一个跟 'pat' 最像？

2. Hausdorff 距离

$$D_H(S_1, S_2) = \max\left(\underbrace{\max_{s_1 \in S_1}\min_{s_2 \in S_2} D(s_1, s_2)}_{\text{Part A}}, \underbrace{\max_{s_2 \in S_2}\min_{s_1 \in S_1} D(s_2, s_1)}_{\text{Part B}}\right)$$

这个公式看起来很绕，其实包含三个步骤（“最小-最大-最大”）：

1. 第一步（找最近）： 对于集合 S_1 中的每一个点，去集合 S_2 里找离它最近的那个点，记录下这个最短距离。
2. 第二步（找最远，即 Part A）： 在第一步记录的所有“最短距离”中，挑出一个最大的。这代表了“ S_1 中最不合群（离S_2 最远）的那个点”的距离。这叫单向 Hausdorff 距离 $h(S_1, S_2)$ 。
3. 第三步(双向比较)： 反过来做一遍（看 S_2中离S_1 最远的点），得到 Part B。最后取 Part A 和 Part B 中的最大值。

Ch5 线性判别函数

5.1 判别函数概念回顾

1. 决策规则 (Decision Rule)：判别 x 属于 ω_i 类，如果 $g_i(x) > g_j(x)$ 对于所有 $j \ne i$
2. 与贝叶斯的关系：
   a. 最小错误率：可令 $g_i(x) = P(\omega_i|x)$
   b. 最小风险：可令 $g_i(x) = -R(\alpha_i|x)$

5.2 线性判别函数

5.2.1 定义

1. 判别函数$g_i(x)$是 x 的线性函数
2. 一般形式 (c 个类别)：
   a. $g_i(x) = w_i^t x + w_{i0}$
   b. w_i ：权值向量 (Weight vector)
   c. w_{i0} ：偏置 (Bias) 或阈值 (Threshold)

5.2.2 两类别情况

1. **简化：** 对于两类别问题（ ω_1, ω_2 ），我们不需要两个函数 $g_1(x)$ 和 $g_2(x)$
2. **单一函数 g(x)：** 我们可以定义一个单独的判别函数 $g(x) = g_1(x) - g_2(x)$
   a. $g(x) = (w_1^t x + w_{10}) - (w_2^t x + w_{20})$
   b. $g(x) = (w_1 - w_2)^t x + (w_{10} - w_{20})$
3. **最终形式：** $g(x) = w^t x + b$

- 其中： $w = w_1 - w_2$ 且 $b = w_{10} - w_{20}$
- 如果 g(x) > 0，则判别为 ω_1 ；否则，判给 ω_2

5.3 求解 w 和 b

1. 任务： 找到一组 w 和 b，使得对于所有 n 个训练样本，$\omega_i \cdot (w^t x_i + b) > 0$ 这个条件都成立

5.4 准则函数

为了找到最优的 (w, b)，我们定义一个准则函数 J(w, b)，然后试图最小化这个函数。

- 这个函数 J 用来衡量我们当前的 w 和 b 有多“差”。
- 如果 w, b 很好地分类了所有数据，J 的值就很低。
- 如果 w, b 犯了很多错误，J 的值就很高。

5.4.2 准则函数示例

1. $J(w,b) = \sum_{i=1}^{n}(g(x_i) - \omega_i)^2 = \sum_{i=1}^{n}(w^t x_i + b - \omega_i)^2$
2. $J(w,b) = -\sum_{i=1}^{n}\omega_i \cdot g(x_i) = -\sum_{i=1}^{n}\omega_i(w^t x_i + b)$

5.4.3 求解线性判别函数的梯度

1. 示例 1： $J(w,b) = \sum_{i=1}^{n}(w^t x_i + b - \omega_i)^2$

$$\nabla J(w,b) = 2 \cdot \sum_{i=1}^{n}(w^t x_i + b - \omega_i) \cdot \begin{bmatrix} x_i \\ 1 \end{bmatrix}$$

2. 示例 2： $J(w,b) = -\sum_{i=1}^{n}\omega_i(w^t x_i + b)$

$$\nabla J(w,b) = -\sum_{i=1}^{n}\omega_i \cdot \begin{bmatrix} x_i \\ 1 \end{bmatrix}$$

Ch6 多层神经网络

6.1 神经网络的基础

6.1.1 人工神经网络 (ANN)

1. 基本架构：一个典型的 ANN 由三部分组成：
   - 输入层 (Input Layer)
   - 隐藏层 (Hidden Layer)
   - 输出层 (Output Layer)
2. w_{ji} : input-to-hidden layer weight (*i-th* feature to *j-th* hidden unit)
3. w_{kj} : hidden-to-output layer weight (*j-th* hidden to *k-th* output unit)

6.1.2 激活函数

1. Sigmoid (Logistic):

$$f(x) = \frac{1}{1+e^{-x}} \qquad f' = f(1-f)$$

2. tanh (双曲正切):

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \qquad f' = 1 - f^2$$

6.2 前馈过程 (Feedforward Procedure)

1. 输入 → 隐藏层：计算隐藏层神经元 j 的净输入 net_j

$$net_j = \sum_{i=1}^{d} w_{ji}x_i$$

2. 隐藏层激活：计算 j 的输出 y_j

$$y_j = f(net_j)$$

3. 隐藏层 → 输出层：计算输出层神经元 k 的净输入 net_k

$$net_k = \sum_{j=1}^{n_H} w_{kj}y_j$$

4. 最终输出：计算网络最终输出 z_k

$$z_k = f(net_k)$$

6.3 反向传播过程 (Backpropagation Procedure)

1. 误差 （δ） 从输出层反向传播到隐藏层。
2. 计算输出层误差 (δ_k)：
   - $\delta_k = (t_k - z_k) \cdot f'(net_k)$。
   - 误差 = (期望 - 实际）激活函数的导数
3. 计算隐藏层误差 (δ_j)：
   - 隐藏层的误差 j 是所有连接到它的输出层单元 k 的误差的加权和，再乘以 j 的激活导数。
   - $\delta_j = f'(net_j) \cdot [\sum_{k=1}^{c} w_{kj}\delta_k]$。
4. 更新权重：
   - 隐藏层 → 输出层： $\Delta w_{kj} = \eta \cdot \delta_k \cdot y_j$
   - 输入层 → 隐藏层： $\Delta w_{ji} = \eta \cdot \delta_j \cdot x_i$

6.4 训练模式

1. 随机 (Stochastic) 训练：每（随机）选择一个样本，就立即计算梯度并更新一次权重。
2. 批量 (Batch) 训练：遍历所有训练样本，累加所有样本产生的梯度，然后在每个 epoch（完整的一轮）结束时更新一次权重。

Ch7 补充

7.1 Applications of Pattern Recognition

1. **Character Recognition：**
   a. **Input:** images with characters (normally contaminated with noise)
   b. **Output:** The identified character strings
   Useful in scenarios such as **automatic license plate recognition (ALPR)**, **optical character recognition(OCR)**, etc.
2. **Speech Recognition：**
   a. **Input：** acoustic signal (e.g. sound waves)
   b. **Output:** contents of the speech
   Useful in scenarios such as **speech-to-text (STT)**, **voice command & control**, etc.
3. **Fingerprint Recognition：**
   a. **Input:** fingerprints of some person
   b. **Output：** the person's identity
   Useful in scenarios such as **computerized access control**, **criminal pursuit**, etc.
4. **Signature Identification：**
   a. **Input:** signature of some person (sequence of dots)
   b. **Output:** the signatory's identity
   Useful in scenarios such as **digital signature verification**, **credit card anti-fraud**, etc.
5. **Face Detection：**
   a. **Input:** images with several people
   b. **Output:** locations of the peoples' faces in the image
   Useful in scenarios such as **digital camera capturing**, **video surveillance**, etc.
6. **Text Categorization：**
   a. **Input:** document, web pages, etc.
   b. **Output:** category of the text, such as political, economic, military, sports, etc.
   Useful in scenarios such as **information retrieval**, **document organization**, etc.

7.2 Basic Concepts

1. **Model (模型)：** Descriptions which are typically mathematical in form
   a. e.g. image → matrix; sound waves→ frequency vector
2. **Sample (样本)：** Representatives of the patterns we want to classify
   a. e.g. fingerprint of a suspect; ECG of a patient
3. **Training Set (训练集)：** A set of samples used to train classifiers
4. **Test Set (测试集)：** A set of samples to be classified, usually being mutually exclusive to training set
5. **Feature (特征)：** Attributes which characterize properties of the samples
6. **Feature Vector (特征向量)：** Vector formed by a group of features, usually in column form[由一组特征组成的向量，通常表示为列向量]
7. **Feature Space (特征空间)：** Space containing all the possible feature vectors(由所有可能的特征向量组成的数据空间)
8. **Scatter Plot (散布图)：** Each sample is plotted as a point in the feature space(将每个样本表示为特征空间中的一个点)
9. **Decision Boundary (决策边界)：** Boundaries in feature space which separate different categories(特征空间中区分各个类别的边界)
10. **Noise (噪声)**
    - Chinese: 数据中随机、无关的干扰信号（如图像噪点），会掩盖真实模式。
    - English: Random, irrelevant interference in data (e.g., image grain) that obscures true patterns.
11. **Segmentation (分割)**
    - Chinese: 将感兴趣的目标（前景）从背景或其他物体中分离出来的过程。
    - English: The process of separating the object of interest (foreground) from the background or other objects.
12. **Data Collection (数据采集)**
    - Chinese: 收集足够数量且具有代表性的样本，用于模型的训练和测试。
    - English: Gathering a sufficient number of representative samples for model training and testing.
13. **Domain Knowledge (领域知识)**
    - Chinese: 利用特定领域的专家经验（而非纯统计）来辅助特征选择或模型设计。
    - English: Using expert experience in a specific field (rather than pure statistics) to assist in feature selection or model design.
14. **Feature Extraction (特征抽取)**
    - Chinese: 从原始数据中提取出最具区分度的关键信息，以降低维度并去除冗余。
    - English: Extracting the most distinctive key information from raw data to reduce dimensionality and remove redundancy.
15. **Pattern Representation (模式表示)**
    - Chinese: 将现实世界的对象转化为计算机可处理的数学形式（通常是特征向量）。
    - English: Converting real-world objects into mathematical forms (usually feature vectors) that computers can process.

16. Model Selection (模型选择)
- Chinese: 根据数据特点，选择最合适的数学模型结构（如选择神经网络还是SVM）。
- English: Choosing the most suitable mathematical model structure (e.g., Neural Networks vs. SVM) based on data characteristics.

17. Overfitting (过配/过拟合)
- Chinese: 模型死记硬背训练数据（包括噪声），导致在遇到新数据时表现很差。
- English: The model "memorizes" training data (including noise), leading to poor performance on new, unseen data.

18. Context (上下文)
- Chinese: 利用周围的信息或依赖关系来辅助识别（例如利用单词拼写规则识别字母）。
- English: Using surrounding information or dependencies to assist recognition (e.g., using spelling rules to identify blurred letters).

19. Classifier Ensemble (分类器集成)
- Chinese: 组合多个不同的分类器共同决策，通常比单个分类器更准、更稳。
- English: Combining multiple different classifiers to make joint decisions, typically more accurate and robust than a single one.

20. Costs and Risks (代价与风险)
- **1. Definitions & Difference (定义与区别)**
  - **Chinese:**
    - 代价 (Cost, λ): 针对某一次特定决策所赋予的惩罚值。例如：把"健康"误判为"患病"，这个单一错误的后果就是代价。
    - 风险 (Risk, ***R***): 代价的数学期望（平均值）。它结合了发生错误的概率和该错误的代价。
    - 核心区别: 代价是单次结果的后果，风险是长期来看的平均预期损失。决策的目标通常是最小化风险，而不仅仅是最小化错误次数。
  - **English:**
    - Cost (λ): The specific penalty assigned to a single decision outcome. E.g., The consequence of misclassifying "Healthy" as "Sick".
    - Risk (***R***): The expected value of the cost. It combines the probability of an error with the cost of that error.
    - Key Difference: Cost refers to the consequence of one event, while Risk is the average expected loss. The goal is to minimize risk, not just the error rate.
- **2. Are they equal? (代价是否相等?)**
  - **Chinese:**
    - 不，通常是不等的 (Unequal Costs)。
    - 在现实应用中，不同类型的错误后果严重程度完全不同。
    - 例子: 在癌症筛查中，漏报（有病没查出来 → 延误治疗导致死亡）的代价远高于误报（没病查出有病 → 虚惊一场和额外的检查费）。因此，系统宁可误报也不能漏报。
  - **English:**
    - No, they are usually unequal.
    - In real-world applications, different types of errors carry vastly different severities.
    - Example: In cancer screening, a False Negative (missing a cancer case → delayed treatment/death) has a much higher cost than a False Positive (false alarm → anxiety and extra tests). Therefore, systems are biased to avoid False Negatives.

21. Computational Complexity (计算复杂度)
- Chinese: 算法运行所需的时间和内存资源。
- English: The time and memory resources required to execute the algorithm.

7.3 Give your descriptions on two real-world applications of pattern recognition techniques. Furthermore, based on your own understanding, for each application please explain: i) What are the domain knowledge which we could utilize to facilitate the pattern recognition process; ii) In case of making incorrect decisions, do they have equal cost or unequal cost ?

Application 1: Character Recognition (License Plate Recognition)
- **Description:**
  - Input: Noisy images of vehicles.
  - Output: The license plate character string.
  - Scenario: Automatic toll collection or traffic law enforcement.
- **i) Domain Knowledge:**
  - Syntax Rules: Utilizing government-mandated formats (e.g., "3 letters, 3 numbers") helps resolve ambiguities (e.g., distinguishing 'O' from '0' based on position).
  - Geometry: Knowing the fixed rectangular aspect ratio of plates helps the algorithm filter out background noise like bumper stickers.
- **ii) Cost of Incorrect Decisions (Unequal):**
  - Unequal. A False Positive (wrongly ticketing an innocent driver) incurs high legal and social costs. A False Negative (missing a speeder) only results in lost revenue. Thus, the cost of a False Positive is higher.

Application 2: Fingerprint Recognition (Forensics)
- **Description**:
  - Input: Latent fingerprints from a crime scene.
  - Output: The identity of the person (matched from a database).
  - Scenario: Criminal identification.
- **i) Domain Knowledge:**
  - Minutiae: Focusing on biological ridge features (endings and bifurcations) rather than comparing raw pixels makes recognition robust.
  - Elasticity: Algorithms use physical models of skin elasticity to compensate for distortion when a finger is pressed/stretched.
- **ii) Cost of Incorrect Decisions (Unequal):**
  - Highly Unequal. A False Match (wrongful conviction of an innocent person) is a catastrophic failure of justice. A False Non-Match (criminal unidentified) delays the investigation but harms no innocent individual. Therefore, False Match costs are significantly higher.

7.4 例题一

Q2. Consider the Bayesian belief network as shown in Figure 1.

1. Which of the following are asserted by the network structure and why?
   (a) $P(B, I, G) = P(B)P(I)P(G)$.
   (b) $P(I|B, L) = P(I|B, L, G)$.
2. Calculate the value of $P(B = f, I = t, L = t, G = f, J = f)$.
3. Calculate the probability that someone will go to jail given that he/she has not broken the law, has been indicted, and his/her lawyer is experienced.

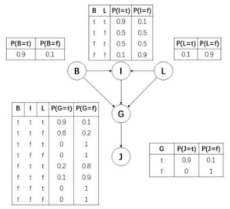
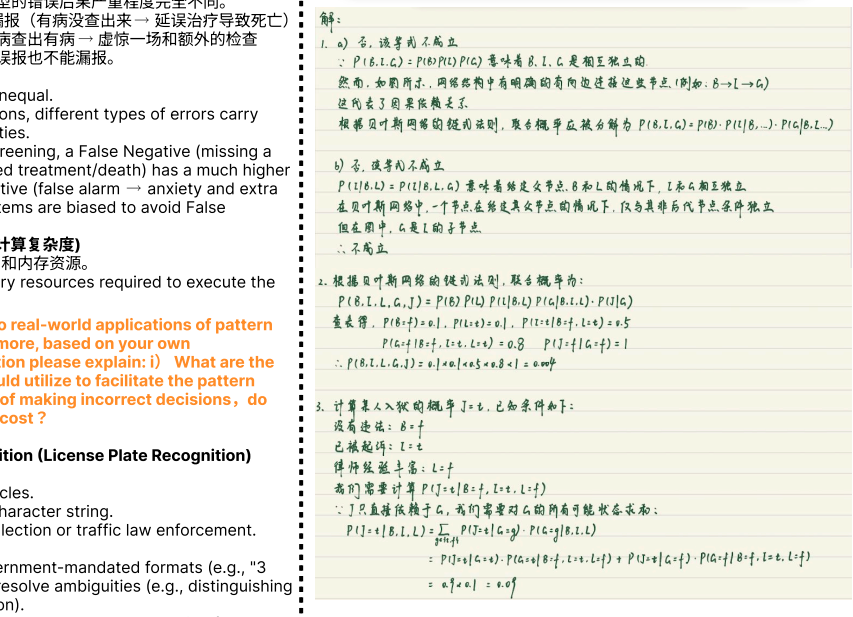


Figure 1: A Bayesian belief network with Bernoulli-distributed variables B = Broke Law, I = Indicted, L = Inexperienced Lawyer, G = Found Guilty, J = Jailed.



7.5 例题二

Q2. Consider a three-layer feedforward neural network with *four* input neurons, *two* hidden neurons, and *one* output neurons, as illustrated in Fig 2.

The activation function from hidden layer to out layer is *Sigmoid* function and the activation function of the output is also the *Sigmoid* function, where *Sigmoid* is $s(x) = \frac{1}{1+e^{-x}}$.

Given the initial model:

$$W^{(0)} = \begin{bmatrix} 0.8 & 0.5 & 0.6 & 0.6 \\ 1.0 & 0.3 & 0.4 & 1.2 \end{bmatrix}, W^{(1)} = \begin{bmatrix} 0.3 \\ 0.6 \end{bmatrix}$$

where $W^{(0)}$ is the weight matrix from input layer to hidden layer and $W^{(1)}$ is the weight matrix from hidden layer to out layer. We use the notation $w_{ij}^{(k)}$ to represent the element at i-th row j-th column of matrix $W^{(k)}$.

We take the MSE (minimum square error) as the loss function:

$$J_w(\mathbf{p}, \mathbf{t}) = \frac{1}{2}\sum_i (\mathbf{p}_i - \mathbf{t}_i)^2$$

Suppose *Stochastic Backpropagation Algorithm* is used to train the neural network, and the randomly chosen training example is $\mathbf{x} = (0, 4, 2, 3)^\top$ with $\mathbf{t} = 1$. What is the learned model $\mathbf{W}$ after training the neural networks (Learning rate $\eta = 1$, round the results to the fourth decimal place).

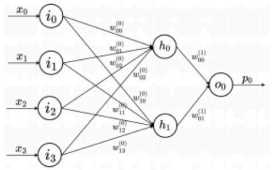
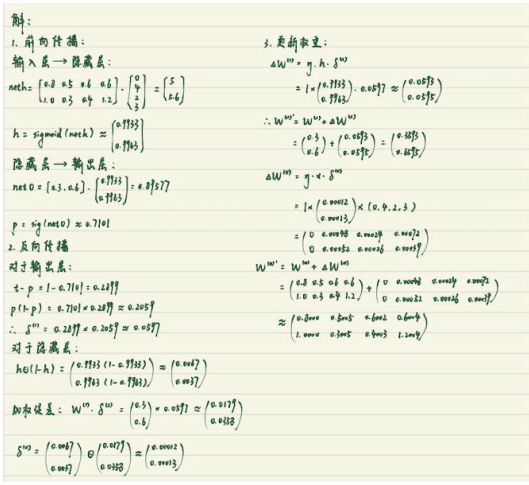
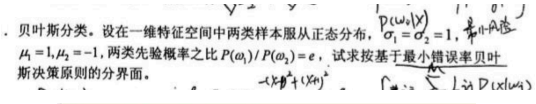
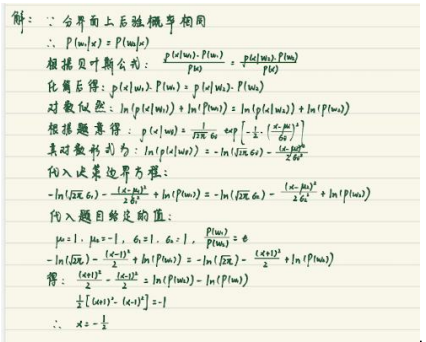


Figure 2: The architecture of a three-layer feedforward neural network.



7.6 例题三





7.7 例题三

Q1. In a two-class three-dimensional classification problem, the feature vectors of each class obey the Gaussian distribution, and the covariance matrix is

$$\boldsymbol{\Sigma}_1 = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \boldsymbol{\Sigma}_2 = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1/4 \end{pmatrix}$$

The mean vectors $\boldsymbol{\mu}$ for each class are $\boldsymbol{\mu}_1 = [1, 4, 2]^\top$ and $\boldsymbol{\mu}_2 = [2, 2, 4]^\top$, respectively. We assume that the prior probabilities $p(\omega_i)$ of the two classes are equal. Please give the formulation of the decision boundary.

Hints: The discriminant function is given by

$$g_i(\mathbf{x}) = \mathbf{x}^\top \mathbf{W}_i \mathbf{x} + \mathbf{w}_i^\top \mathbf{x} + w_{i0}$$

where

$$\mathbf{W}_i = -\frac{1}{2}\boldsymbol{\Sigma}_i^{-1}$$

$$\mathbf{w}_i = \boldsymbol{\Sigma}_i^{-1}\boldsymbol{\mu}_i$$

and

$$w_{i0} = -\frac{1}{2}\boldsymbol{\mu}_i^\top \boldsymbol{\Sigma}_i^{-1}\boldsymbol{\mu}_i - \frac{1}{2}\ln|\boldsymbol{\Sigma}_i| + \ln p(\omega_i)$$

7.8 期中考试第一题

Q1. In a two-class three-dimensional classification problem, the feature vectors of each class obey the Gaussian distribution, and the covariance matrix is

$$\boldsymbol{\Sigma}_1 = \begin{pmatrix} 1/6 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \boldsymbol{\Sigma}_2 = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1/10 \end{pmatrix}$$

The mean vectors $\boldsymbol{\mu}$ for each class are $\boldsymbol{\mu}_1 = [2, 4, 5]^\top$ and $\boldsymbol{\mu}_2 = [1, 2, -4]^\top$, respectively. We assume that the prior probabilities $p(\omega_i)$ of the two classes are equal. Please give the formulation of the decision boundary.

Hints: The discriminant function is given by

$$g_i(\mathbf{x}) = \mathbf{x}^\top \mathbf{W}_i \mathbf{x} + \mathbf{w}_i^\top \mathbf{x} + w_{i0}$$

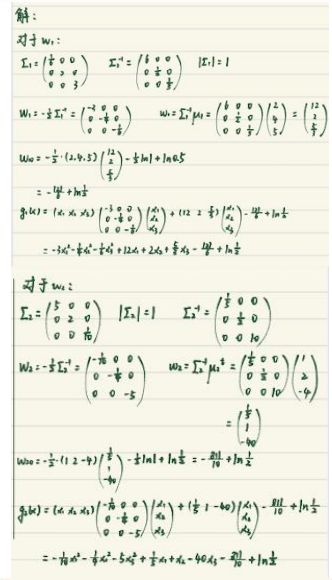
where

$$\mathbf{W}_i = -\frac{1}{2}\boldsymbol{\Sigma}_i^{-1}$$

$$\mathbf{w}_i = \boldsymbol{\Sigma}_i^{-1}\boldsymbol{\mu}_i$$

and

$$w_{i0} = -\frac{1}{2}\boldsymbol{\mu}_i^\top \boldsymbol{\Sigma}_i^{-1}\boldsymbol{\mu}_i - \frac{1}{2}\ln|\boldsymbol{\Sigma}_i| + \ln p(\omega_i)$$



Q1. Perceptron Algorithms

Suppose we have four normalized training samples: $y_1(1,4,1)^T, y_2(1,4,2)^T, y_3(-1,0,-1)^T, y_4(-1,-1,-1)^T$. The discriminant function $g(\boldsymbol{y}) = \boldsymbol{a}^T\boldsymbol{y}$ is used to minimize the criterion function $J_p(\boldsymbol{a}) = \sum_\gamma(-\boldsymbol{a}^T\boldsymbol{y})$, where γ denotes the misclassified set ($a^T y_i \leqslant 0$).

(a) Fixed-Increment Single-Sample Correction

- **Params:** $a(0) = (-2,0,0)^T, \eta = 1$.
- **Rule:** $a^T y_k \leqslant 0 \implies a \leftarrow a + \eta \cdot y_k$.

Round 1

- y_1: $a^T y_1 = -2 \leqslant 0$ (**E**). $a \leftarrow (-2,0,0)+1\cdot y_1 = (\mathbf{-1,4,1})^\mathbf{T}$.
- y_2: $a^T y_2 = 17 > 0$ (P).
- y_3: $a^T y_3 = 0 \leqslant 0$ (**E**). $a \leftarrow (-1,4,1)+1\cdot y_3 = (\mathbf{-2,4,0})^\mathbf{T}$.
- y_4: $a^T y_4 = -2 \leqslant 0$ (**E**). $a \leftarrow (-2,4,0)+1\cdot y_4 = (\mathbf{-3,3,-1})^\mathbf{T}$.

Round 2

- y_1: $a^T y_1 = 8 > 0$ (P). y_2: $a^T y_2 = 7 > 0$ (P).
- y_3: $a^T y_3 = 4 > 0$ (P). y_4: $a^T y_4 = 1 > 0$ (P).

Conclusion (a): Algorithm terminates.

$$\boxed{g(y) = -3x_1 + 3x_2 - x_3}$$

(b) Batch Perceptron Algorithm

- **Params:** $a(0) = (-2,-1,1)^T, \eta = 0.5, \theta = 0.5$.
- **Stop Condition:** $\|a(k+1) - a(k)\| \leqslant 0$ or no error samples.
- **Rule:** Err if $a^T y \leqslant 0.5$. $a \leftarrow a + \eta \sum_{\text{err}} y$.

Round 1 $a = (-2,-1,1)^T$. Check all samples:

- $y_1 = -5$ (**E**); $y_2 = -4$ (**E**). y_3, y_4 (P).
- $\sum_{err} y = (2,8,3)^T$. $a(1) = a+0.5(2,8,3) = (\mathbf{-1,3,2.5})^\mathbf{T}$.

Round 2 $a = (-1,3,2.5)^T$. Check all samples:

- y_3: -1.5 (**E**); y_4: -4.5 (**E**). y_1, y_2 (P).
- $\sum_{err} y = (-2,-1,-2)^T$. $a(2) = a + 0.5(-2,-1,-2) = (\mathbf{-2,2.5,1.5})^\mathbf{T}$.

Round 3 $a = (-2,2.5,1.5)^T$. Check all samples:

- y_3: 0.5 (Boundary/Pass). y_4: -2 (**E**).
- $a(3) = a(2) + 0.5y_4 = (\mathbf{-2.5,2,1})^\mathbf{T}$.

Final Result (b): $a_{final} = (\mathbf{-3,1.5,0.5})^\mathbf{T}$.

$$\boxed{g(y) = -3x_1 + 1.5x_2 + 0.5x_3}$$

Q2. Bayesian Inference: $p(x \mid \theta) \sim U(0,\theta)$

Setup: Prior $p(\theta) \sim U(1,9)$. Samples $D^4 = \{2,4,6,8\}$. **Goal:** Find $p(x \mid D^4)$.

1. Likelihood and Posterior $p(\theta \mid D^4)$:

- **Prior:** $p(\theta) = \frac{1}{8}$, for $1 \leqslant \theta \leqslant 9$, 0 otherwise.
- **Likelihood:** $p(D^4 \mid \theta) = \theta^{-4}$ only if $\theta \geqslant 8$.
- **Posterior** $p(\theta \mid D^4) \propto p(\theta) \times p(D^4 \mid \theta) = \alpha \times \theta^{-4}$ **on** $\theta \in [8,9]$.
- **Normalization (**α**):** $\int_8^9 \alpha\theta^{-4}\,d\theta = 1$, so $\alpha = \left[-\frac{1}{3}\theta^{-3}\right]_8^9 = \frac{1}{3}\left(8^{-3} - 9^{-3}\right)$.

2. Predictive Distribution

We have the posterior:

$$p(\theta \mid D^4) = \begin{cases} \alpha\,\theta^{-4}, & 8 \leqslant \theta \leqslant 9, \\ 0, & \text{otherwise.} \end{cases}$$

We now compute the posterior predictive distribution:

$$p(x \mid D^4) = \int p(x \mid \theta)\, p(\theta \mid D^4)\, d\theta = \int_8^9 \frac{1}{\theta}\cdot\alpha\theta^{-4}\, d\theta = \alpha\int_8^9 \theta^{-5}\, d\theta.$$

Case Analysis

Case A: $0 \leqslant x \leqslant 8$

Since $x \leqslant \theta$ holds for all $\theta \in [8,9]$,

$$p(x \mid D^4) = \alpha\int_8^9 \theta^{-5} d\theta = \frac{\alpha}{4}\left(8^{-4} - 9^{-4}\right).$$

Case B: $8 < x \leqslant 9$

Here only $\theta \geqslant x$ contributes:

$$p(x \mid D^4) = \alpha\int_x^9 \theta^{-5} d\theta = \frac{\alpha}{4}\left(x^{-4} - 9^{-4}\right).$$

Case C: $x < 0$ or $x > 9$

$$p(x \mid D^4) = 0.$$

Final Result

$$p(x \mid D^4) = \begin{cases} \frac{\alpha}{4}\left(8^{-4} - 9^{-4}\right), & 0 \leqslant x \leqslant 8, \\ \frac{\alpha}{4}\left(x^{-4} - 9^{-4}\right), & 8 < x \leqslant 9, \\ 0, & \text{otherwise.} \end{cases}$$

Q3: Three-Class Classification Using Pairwise ω_i/ω_j Method We consider a three-class classification problem with discriminant functions:

$$d_1(x) = x_1 + 2x_2 - 4,$$
$$d_2(x) = x_1 - 4x_2 + 4,$$
$$d_3(x) = -x_1 + 3.$$

The goal is to apply the pairwise "ω_i/ω_j" (one-vs-one) strategy to classify the feature space and determine the class of the pattern:

$$X = \begin{bmatrix} 5 \\ 2 \end{bmatrix}.$$

1. One-vs-One Method Explanation

For each class pair (ω_i, ω_j), build a binary classifier.

With $C = 3$ classes, the number of classifiers is:

$$\frac{C(C-1)}{2} = 3.$$

- Classifier for ω_1 vs ω_2
- Classifier for ω_1 vs ω_3
- Classifier for ω_2 vs ω_3

Decision rule: ω_i wins if $d_i(x) > d_j(x)$, ω_j wins otherwise.
Each classifier partitions the (x_1, x_2)-plane along:

$$d_i(x) = d_j(x).$$

Each classifier gives one vote to the winning class.

2. Determine the Class of $X = [5,2]^T$

Step 1: Compute discriminant values

$$d_1(5,2) = 5 + 2\cdot 2 - 4 = 5,$$
$$d_2(5,2) = 5 - 4\cdot 2 + 4 = 1,$$
$$d_3(5,2) = -5 + 3 = -2.$$

Step 2: Pairwise voting

- ω_1 **vs** ω_2: $5 > 1 \Rightarrow \omega_1$ gets 1 vote.
- ω_1 **vs** ω_3: $5 > -2 \Rightarrow \omega_1$ gets 1 vote.
- ω_2 **vs** ω_3: $1 > -2 \Rightarrow \omega_2$ gets 1 vote.

Step 3: Total votesω_1 : 2 votes, ω_2 : 1 vote, ω_3 : 0 votes.
Final Conclusion

ω_1 receives the highest number of votes.

Therefore,

$X = [5,2]^T$ belongs to class ω_1.

Q4: Bayesian Classification and Decision Theory

Suppose we have two categories ω_1 and ω_2, and the one-dimensional training examples for each category are:

$$D_1 = \{-6,-4,-3,-1,0,2\},$$
$$D_2 = \{-2,1,4,5\}.$$

Please answer the following questions.

(a) Bayes Theorem (6 points)

The Bayes Theorem is named after Thomas Bayes (1702–1761). What is his **nationality** and **occupation**?

(b) Maximum-Likelihood Estimation (10 points)

Assume the likelihood of each category is Gaussian:

$$p(x \mid \omega_1) \sim \mathcal{N}(\mu_1, \sigma_1^2), \qquad p(x \mid \omega_2) \sim \mathcal{N}(\mu_2, \sigma_2^2).$$

Using ML estimates of μ_i, σ_i^2, **which category should we choose for** $x = -0.5$?

(c) Decision with Loss Function (8 points)

Based on the ML estimates above, suppose we have three actions

$$A = \{\alpha_1, \alpha_2, \alpha_3\},$$

with the following loss matrix:

category	α_1	α_2	α_3
ω_1	5	10	20
ω_2	10	30	3

For $x = -0.5$, **which action should be taken**?

(d) Bayesian Estimator (8 points)

Assume now that:

$$\sigma_1 = 1, \quad \sigma_2 = 2,$$
$$\mu_1 \sim \mathcal{N}(-2,1), \qquad \mu_2 \sim \mathcal{N}(2,1).$$

Under Bayesian estimation, **which category should we select for** $x = -0.5$?

(e) Nonparametric k-NN Classification (8 points)

Assume the likelihood does not have a parametric form. Using **k-nearest neighbors** with $k = 2$, **which category should we assign to** $x = -0.5$?

Assume the cell grows uniformly around $x = -0.5$.

(a) Thomas Bayes (c. 1701–1761), an 18th-century British mathematician.

(b) Maximum–Likelihood Estimation

Assume the class–conditional likelihoods are Gaussian:

$$p(x \mid \omega_i) = \frac{1}{\sqrt{2\pi\sigma_i^2}}\exp\left[-\frac{(x-\mu_i)^2}{2\sigma_i^2}\right].$$

For a dataset $D_i = \{x_1, \ldots, x_n\}$, the log-likelihood is

$$\ln p(D_i \mid \mu_i, \sigma_i^2) = -\frac{n}{2}\ln\left(2\pi\sigma_i^2\right) - \frac{1}{2\sigma_i^2}\sum_{k=1}^n (x_k - \mu_i)^2.$$

Maximizing w.r.t. μ_i and σ_i^2:

$$\hat{\mu}_i = \frac{1}{n}\sum_{k=1}^n x_k, \qquad \hat{\sigma}_i^2 = \frac{1}{n}\sum_{k=1}^n (x_k - \hat{\mu}_i)^2.$$

Class ω_1:

$$D_1 = \{-6,-4,-3,-1,0,2\}, \qquad n = 6.$$

Mean:

$$\hat{\mu}_1 = \frac{-6-4-3-1+0+2}{6} = \frac{-12}{6} = -2.$$

Variance:

$$\hat{\sigma}_1^2 = \frac{1}{6}\left[(-6+2)^2 + (-4+2)^2 + (-3+2)^2 + (-1+2)^2 + 0^2 + (2+2)^2\right] = 7$$

Likelihood at $x = -0.5$:

$$p(x = -0.5 \mid \omega_1) = \frac{1}{\sqrt{2\pi\times 7}}\exp\left[-\frac{(-0.5+2)^2}{2\times 7}\right] \approx 0.128.$$

Posterior (equal priors):

$$P(\omega_1 \mid x = -0.5) \propto p(x \mid \omega_1) \times p(\omega_1) = 0.077.$$

Class ω_2:

$$D_2 = \{-2,1,4,5\}, \qquad n = 4.$$

Mean:

$$\hat{\mu}_2 = \frac{-2+1+4+5}{4} = \frac{8}{4} = 2.$$

Variance:

$$\hat{\sigma}_2^2 = \frac{1}{4}\left[(-2-2)^2 + (1-2)^2 + (4-2)^2 + (5-2)^2\right] = \frac{30}{4} = 7.5.$$

Likelihood at $x = -0.5$:

$$p(x = -0.5 \mid \omega_2) = \frac{1}{\sqrt{2\pi\times(7.5)}}\exp\left[-\frac{(-0.5-2)^2}{2\times(7.5)}\right] \approx 0.0960.$$

Posterior (equal priors):

$$P(\omega_2 \mid x = -0.5) \propto p(x \mid \omega_2) \times p(\omega_2) = 0.0384.$$

Decision for x=-0.5:

$$\frac{P(\omega_1 \mid x = -0.5)}{P(\omega_2 \mid x = -0.5)} > \omega_1$$

$$\boxed{\text{Choose } \omega_1.}$$

(c) Decision with Loss Function

We are given the loss matrix:

category	α_1	α_2	α_3
ω_1	5	10	20
ω_2	10	30	3

From part (b), the (unnormalized) posterior values at $x = -0.5$ are:

$$P(\omega_1 \mid x) \propto 0.077, \qquad P(\omega_2 \mid x) \propto 0.0384.$$

Since the denominator $p(x)$ is common to all actions, we compare only numerators.

Risk of action α_1:

$$R(\alpha_1 \mid x = -0.5) = \lambda(\alpha_1|\omega_1)P(\omega_1|x) + \lambda(\alpha_1|\omega_2)P(\omega_2|x)$$
$$= 5(0.077) + 10(0.0384) = \boxed{0.769}.$$

Risk of action α_2:

$$R(\alpha_2 \mid x = -0.5) = 10(0.077) + 30(0.0384) = \boxed{1.922}.$$

Risk of action α_3:

$$R(\alpha_3 \mid x = -0.5) = 20(0.077) + 3(0.0384) = \boxed{1.6522}.$$

Conclusion:

$$R(\alpha_1) < R(\alpha_3) < R(\alpha_2)$$

$$\boxed{\text{Choose action } \alpha_1.}$$

(d) Bayesian Estimator

Assume

$$\sigma_1 = 1, \qquad \sigma_2 = 2,$$

and priors

$\mu_1 \sim \mathcal{N}(\mu_{01}, \sigma_{01}^2) = \mathcal{N}(-2,1), \qquad \mu_2 \sim \mathcal{N}(\mu_{02}, \sigma_{02}^2) = \mathcal{N}(2,1).$

Given data:

$$D_1 = \{-6,-4,-3,-1,0,2\}, \quad n_1 = 6, \quad \bar{x}_1 = -2,$$

$$D_2 = \{-2,1,4,5\}, \quad n_2 = 4, \quad \bar{x}_2 = 2.$$

For a Gaussian prior and Gaussian likelihood, the posterior variance and mean are

$$\frac{1}{\sigma_{n,i}^2} = \frac{1}{\sigma_i^2} + \frac{n_i}{\sigma_{0i}^2}, \qquad \mu_{n,i} = \sigma_{n,i}^2\left(\frac{n_i\bar{x}_i}{\sigma_i^2} + \frac{\mu_{0i}}{\sigma_{0i}^2}\right).$$

Category ω_1:

$$\sigma_1^2 = 1, \quad \sigma_{01}^2 = 1, \quad n_1 = 6, \quad \bar{x}_1 = -2, \quad \mu_{01} = -2.$$

Posterior variance:

$$\frac{1}{\sigma_{n,1}^2} = \frac{1}{\sigma_{01}^2} + \frac{n_1}{\sigma_1^2} = 1 + 6 = 7 \quad \Rightarrow \quad \sigma_{n,1}^2 = \frac{1}{7}.$$

Posterior mean:

$$\mu_{n,1} = \sigma_{n,1}^2 \times (\frac{n_1 \bar{x}_1}{\sigma_1^2} + \frac{\mu_{01}}{\sigma_{01}^2}) = \frac{1}{7}(\frac{6 \cdot (-2)}{1} + \frac{(-2)}{1}) = -2.$$

Thus

$$p(x \mid D_1) = \mathcal{N}\big(-2,\ 1 + \tfrac{1}{7}\big) = \mathcal{N}\big(-2,\ \tfrac{8}{7}\big),$$

so

$$p(x = -0.5 \mid D_1) \approx 0.139.$$

Category ω_2:

$$\sigma_2^2 = 4, \quad \sigma_{02}^2 = 1, \quad n_2 = 4, \quad \bar{x}_2 = 2, \quad \mu_{02} = 2.$$

Posterior variance:

$$\frac{1}{\sigma_{n,2}^2} = \frac{1}{\sigma_{02}^2} + \frac{n_2}{\sigma_2^2} = \frac{1}{1} + \frac{4}{4} = 2 \quad \Rightarrow \quad \sigma_{n,2}^2 = \frac{1}{2}.$$

Posterior mean:

$$\mu_{n,2} = \sigma_{n,2}^2 \times (\frac{n_2 \bar{x}_2}{\sigma_2^2} + \frac{\mu_{02}}{\sigma_{02}^2}) = \frac{1}{2}\left(\frac{4 \cdot 2}{4} + \frac{2}{1}\right) = \frac{1}{2}(4) = 2.$$

Thus

$$p(x \mid D_2) = \mathcal{N}\big(2,\ 4 + \tfrac{1}{2}\big) = \mathcal{N}(2, 4.5),$$

so

$$p(x = -0.5 \mid D_2) \approx 0.094.$$

Decision: with equal priors,

$P(\omega_1 \mid x = -0.5) \propto 0.6 \times 0.139, \qquad P(\omega_2 \mid x = -0.5) \propto \times 0.4 \times 0.024,$

hence

$$\frac{P(\omega_1 \mid x = -0.5)}{P(\omega_2 \mid x = -0.5)} > \omega_1$$

choose ω_1.

(e) Nonparametric k-Nearest Neighbors Classification

Given:

$$x = -0.5, \qquad k = 2,$$

and category sample sizes:

$$N_1 = 6, \qquad N_2 = 4.$$

We determine the class of x using the k nearest neighbors (uniform window).

Step 1. Distances to samples in ω_1

$$\begin{aligned} |-6-(-0.5)| &= 5.5, \\ |-4-(-0.5)| &= 3.5, \\ |-3-(-0.5)| &= 2.5, \\ |-1-(-0.5)| &= 0.5 \quad \text{(nearest)}, \\ |0-(-0.5)| &= 0.5 \quad \text{(nearest)}, \\ |2-(-0.5)| &= 2.5. \end{aligned}$$

The two nearest neighbors are at distance 0.5. Thus:

$$r_1 = 0.5, \qquad V_1 = 2r_1 = 1.$$

Step 2. Distances to samples in ω_2

$$\begin{aligned} |-2-(-0.5)| &= 1.5 \quad \text{(nearest)}, \\ |1-(-0.5)| &= 1.5 \quad \text{(nearest)}, \\ |4-(-0.5)| &= 4.5, \\ |5-(-0.5)| &= 5.5. \end{aligned}$$

The two nearest neighbors occur at distance 1.5. Thus:

$$r_2 = 1.5, \qquad V_2 = 2r_2 = 3.$$

Step 3. Density estimates

Using

$$p(\omega_i \mid x) \propto \frac{k}{N_i V_i},$$

we have:

$$p(\omega_1 \mid x) \propto \frac{2}{6 \cdot 1} = \frac{1}{3},$$

$$p(\omega_2 \mid x) \propto \frac{2}{4 \cdot 3} = \frac{1}{6}.$$

Finally, comparing posteriors using the k-NN density estimates:

$$\frac{P(\omega_1 \mid x = -0.5)}{P(\omega_2 \mid x = -0.5)} = \frac{p_n(x)\, P(\omega_1)}{p_n(x)\, P(\omega_2)} = \frac{\frac{1}{3} \cdot \frac{3}{5}}{\frac{1}{6} \cdot \frac{2}{5}} = 3 > 1.$$

Thus,

Choose ω_1.

Q5. Three-Layer Feedforward Neural Network (Tanh Activation)

Consider a three-layer feedforward neural network with **two** input neurons, **three** hidden neurons, and **two** output neurons, as illustrated in Fig. 1.

The activation function from hidden layer to output layer is *Tanh*, and the activation for the output is also Tanh, where

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

Given the initial model:

$$W^{(0)} = \begin{bmatrix} -0.3 & 0.5 \\ 0.1 & -0.3 \\ 0.1 & 0.2 \end{bmatrix}, \qquad W^{(1)} = \begin{bmatrix} 0.5 & 0.7 & -0.2 \\ 0.3 & -0.6 & 0.3 \end{bmatrix}.$$

Here $W^{(0)}$ is the weight matrix from input layer to hidden layer and $W^{(1)}$ is the weight matrix from hidden layer to output layer. We use the notation $w_{ij}^{(k)}$ to represent the element at the i-th row and j-th column of matrix $W^{(k)}$.

We take the MSE (mean square error) as the loss function:

$$J_w(\mathbf{p}, \mathbf{t}) = \frac{1}{2} \sum_i (p_i - t_i)^2,$$

where $\mathbf{p}$ is the output vector of the network and $\mathbf{t}$ is the labels of given inputs.

Suppose the *Stochastic Backpropagation Algorithm* is used to train the neural network, and the randomly chosen training example is

$$\mathbf{x} = (1, 1)^\top \quad \text{with} \quad \mathbf{t} = (2, 0)^\top.$$

What is the learned model W after training the neural network? (Use learning rate $\eta = 1$, and round all results to the **fourth decimal place**.)

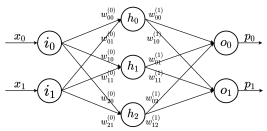


Fig. 1: The architecture of a three-layer feedforward neural network.

Solution to Q5

Given network

$$W^{(0)} = \begin{bmatrix} -0.3 & 0.5 \\ 0.1 & -0.3 \\ 0.1 & 0.2 \end{bmatrix}, \qquad W^{(1)} = \begin{bmatrix} 0.5 & 0.7 & -0.2 \\ 0.3 & -0.6 & 0.3 \end{bmatrix},$$

$$\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \mathbf{t} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \qquad \eta = 1.$$

Activation function is: $\tanh(\cdot)$, Its derivative is $1 - \tanh^2(\cdot)$.

1. Forward propagation

Input layer → hidden layer:

$$\text{net}^{(h)} = W^{(0)}\mathbf{x} = \begin{bmatrix} -0.3 & 0.5 \\ 0.1 & -0.3 \\ -0.1 & 0.2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.3 + 0.5 \\ 0.1 - 0.3 \\ -0.1 + 0.2 \end{bmatrix} = \begin{bmatrix} 0.2 \\ -0.2 \\ 0.1 \end{bmatrix}.$$

$$\mathbf{h} = \tanh\big(\text{net}^{(h)}\big) = \begin{bmatrix} \tanh(0.2) \\ \tanh(-0.2) \\ \tanh(0.1) \end{bmatrix} \approx \begin{bmatrix} 0.1974 \\ -0.1974 \\ 0.0997 \end{bmatrix}.$$

Hidden layer → output layer:

$$\text{net}^{(o)} = W^{(1)}\mathbf{h} = \begin{bmatrix} 0.5 & 0.7 & -0.2 \\ 0.3 & -0.6 & 0.3 \end{bmatrix} \begin{bmatrix} 0.1974 \\ -0.1974 \\ 0.0997 \end{bmatrix} \approx \begin{bmatrix} -0.0593 \\ 0.2046 \end{bmatrix}.$$

$$\mathbf{p} = \tanh\big(\text{net}^{(o)}\big) \approx \begin{bmatrix} -0.0597 \\ 0.2046 \end{bmatrix}.$$

2. Backpropagation

Output layer error:

$$\mathbf{t} - \mathbf{p} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} - \begin{bmatrix} -0.0597 \\ 0.2046 \end{bmatrix} = \begin{bmatrix} 2.0593 \\ -0.2046 \end{bmatrix}.$$

$$1 - \mathbf{p}^2 \approx \begin{bmatrix} 1 - (-0.0597)^2 \\ 1 - (0.2046)^2 \end{bmatrix} \approx \begin{bmatrix} 0.9965 \\ 0.9581 \end{bmatrix}.$$

$$\delta^{(1)} = (\mathbf{t} - \mathbf{p}) \odot (1 - \mathbf{p}^2) \approx \begin{bmatrix} 2.0521 \\ -0.1960 \end{bmatrix}.$$

Hidden layer error:

$$(W^{(1)})^\top \delta^{(1)} = \begin{bmatrix} 0.5 & 0.3 \\ 0.7 & -0.6 \\ -0.2 & 0.3 \end{bmatrix} \begin{bmatrix} 2.0521 \\ -0.1960 \end{bmatrix} \approx \begin{bmatrix} 0.9673 \\ 1.5541 \\ -0.4692 \end{bmatrix}.$$

$$1 - \mathbf{h}^2 \approx \begin{bmatrix} 1 - (0.1974)^2 \\ 1 - (-0.1974)^2 \\ 1 - (0.0997)^2 \end{bmatrix} \approx \begin{bmatrix} 0.9610 \\ 0.9610 \\ 0.9901 \end{bmatrix}.$$

$$\delta^{(0)} = (W^{(1)})^\top \delta^{(1)} \odot (1 - \mathbf{h}^2) \approx \begin{bmatrix} 0.9296 \\ 1.4935 \\ -0.4646 \end{bmatrix}.$$

3. Weight updates ($\eta = 1$)

Output layer weights:

$$\Delta W^{(1)} = \eta\, \delta^{(1)} \mathbf{h}^\top.$$

$$\delta^{(1)} \approx \begin{bmatrix} 2.0521 \\ -0.1960 \end{bmatrix}, \qquad \mathbf{h}^\top \approx \begin{bmatrix} 0.1974 & -0.1974 & 0.0997 \end{bmatrix}.$$

$$\Delta W^{(1)} \approx \begin{bmatrix} 0.4051 & -0.4051 & 0.2046 \\ -0.0387 & 0.0387 & -0.0195 \end{bmatrix}.$$

$$W_{\text{new}}^{(1)} = W^{(1)} + \Delta W^{(1)} \approx \begin{bmatrix} 0.9051 & 0.2949 & 0.0046 \\ 0.2613 & -0.5613 & 0.2805 \end{bmatrix}.$$

Hidden layer weights:

$$\Delta W^{(0)} = \eta\, \delta^{(0)} \mathbf{x}^\top.$$

$$\delta^{(0)} \approx \begin{bmatrix} 0.9296 \\ 1.4935 \\ -0.4646 \end{bmatrix}, \qquad \mathbf{x}^\top = \begin{bmatrix} 1 & 1 \end{bmatrix}.$$

$$\Delta W^{(0)} = \begin{bmatrix} 0.9296 & 0.9296 \\ 1.4935 & 1.4935 \\ -0.4646 & -0.4646 \end{bmatrix}.$$

$$W_{\text{new}}^{(0)} = W^{(0)} + \Delta W^{(0)} \approx \begin{bmatrix} 0.6296 & 1.4296 \\ 1.5935 & 1.1935 \\ -0.5646 & -0.2646 \end{bmatrix}.$$

Final learned model:

$$W_{\text{new}}^{(0)} = \begin{bmatrix} 0.6296 & 1.4296 \\ 1.5935 & 1.1935 \\ -0.5646 & -0.2646 \end{bmatrix},$$

$$W_{\text{new}}^{(1)} = \begin{bmatrix} 0.9051 & 0.2949 & 0.0046 \\ 0.2613 & -0.5613 & 0.2805 \end{bmatrix}$$

Q6. Milk Quality Inspection

Imagine you are a food safety inspector. The supermarket sources milk from two dairy farms, farm A and farm B, with probabilities

$$P(A) = 0.8, \qquad P(B) = 0.2.$$

The probability that milk from farm A is of high quality is 80%, while for farm B the probability is 90%.

You randomly select a carton of milk and find that it is of *low* quality. Based on this information, answer the following questions:

1. Determine from which farm the milk is more likely to have originated, and explain your reasoning.
2. The loss function is defined as $\lambda_{11} = 4$, $\lambda_{12} = 2$, $\lambda_{21} = 2$, $\lambda_{22} = 3$. Using the minimum–risk Bayesian decision rule, determine from which farm the supermarket is more likely to have purchased the milk.

Solution

(a) Let the classes be the farms:

$$\omega = \{A, B\}, \qquad x = \{\text{“low quality”, “high quality”}\}.$$

From the problem:

$$\begin{aligned} P(A) = 0.8, &\qquad P(B) = 0.2, \\ P(\text{high} \mid A) = 0.8 \Rightarrow P(\text{low} \mid A) &= 0.2, \\ P(\text{high} \mid B) = 0.9 \Rightarrow P(\text{low} \mid B) &= 0.1. \end{aligned}$$

Let x be equal to low quality. We want to compute the posterior probabilities:

$$P(A \mid x), \qquad P(B \mid x),$$

using Bayes rule:

$$P(\omega_i \mid x) = \frac{P(\omega_i)\, P(x \mid \omega_i)}{P(x)}$$

Thus,

$$P(x \mid A) = 0.2, \qquad P(x \mid B) = 0.1.$$

Compute posteriors (unnormalized):

$$\begin{aligned} P(A \mid x) \propto P(A)P(x \mid A) &= 0.8 \times 0.2 = 0.16, \\ P(B \mid x) \propto P(B)P(x \mid B) &= 0.2 \times 0.1 = 0.02. \end{aligned}$$

So,

$$P(A \mid x) > P(B \mid x).$$

The milk is more likely from farm A.

(b) Minimum–risk decision

Assumptions:

$$\alpha 1: \text{ Decision is "from Farm A"},$$
$$\alpha 2: \text{ Decision is "from Farm B"}.$$

Loss matrix:

$$\lambda = \begin{pmatrix} \lambda_{11} = 4 & \lambda_{12} = 2 \\ \lambda_{21} = 2 & \lambda_{22} = 3 \end{pmatrix}.$$

Decision rule: choose the action with the smaller risk.

Risk if we decide “from farm A”:

$$\begin{aligned} R(\alpha_A \mid x) &= \lambda_{11} P(A \mid x) + \lambda_{12} P(B \mid x) \\ &\propto 4(0.16) + 2(0.02) \\ &= 0.64 + 0.04 = 0.68. \end{aligned}$$

Risk if we decide “from farm B”:

$$R(\alpha_B \mid x) = \lambda_{21} P(A \mid x) + \lambda_{22} P(B \mid x)$$
$$\propto 2(0.16) + 3(0.02)$$
$$= 0.32 + 0.06 = 0.38.$$

Compare risks:

$$R(\alpha_B) < R(\alpha_A).$$

Minimum-risk decision: choose farm B.

Q7. Bayesian Decision for Supplier Quality

"A new medical test is designed to quickly detect whether someone is infected by SARS or not. The test result could be either positive or negative. A group of 1250 persons have gone through the test among which 1175 are free of SARS infection. Among the 1175 healthy people, 1170 of them return negative test results; while among the remaining infected people, 73 of them return positive test results."

Based on the above information, if *positive* test result is returned for another person, is he/she infected by SARS or not?

From the problem statement, let

$$\omega_1 : \text{No SARS}, \qquad \omega_2 : \text{SARS}, \qquad x \in \{+, -\}.$$

$$P(\omega_1) = \frac{1175}{1250} \approx 0.94, \qquad P(\omega_2) = 1 - P(\omega_1) \approx 0.06.$$

$$P(- \mid \omega_1) = \frac{1170}{1175} \approx 0.9957, \qquad P(+ \mid \omega_2) = \frac{73}{75} \approx 0.9733.$$

$$\therefore P(\omega_1 \mid +) = \frac{P(\omega_1) \cdot P(+ \mid \omega_1)}{P(+)}$$

$$P(\omega_2 \mid +) = \frac{P(\omega_2) \cdot P(+ \mid \omega_2)}{P(+)}.$$

$$P(+ \mid \omega_1) = 1 - P(- \mid \omega_1) = 0.0043,$$

$$P(- \mid \omega_2) = 1 - P(+ \mid \omega_2) = 0.0267.$$

$$\therefore P(+) = P(\omega_1)P(+ \mid \omega_1) + P(\omega_2)P(+ \mid \omega_2)$$

$$= 0.94 \cdot 0.0043 + 0.06 \cdot 0.9733 = 0.06244.$$

$$P(-) = P(\omega_1)P(- \mid \omega_1) + P(\omega_2)P(- \mid \omega_2)$$

$$= 0.94 \cdot 0.9957 + 0.06 \cdot 0.0267 = 0.93756.$$

$$P(\omega_1 \mid +) = \frac{0.98 \cdot 0.0043}{0.02644} \approx 0.0647.$$

$$P(\omega_2 \mid +) = \frac{0.02 \cdot 0.9733}{0.02644} \approx 0.9353.$$

$$\therefore P(\omega_2 \mid +) > P(\omega_1 \mid +).$$

$Conclusion: Infected\ with\ SARS.$

Q8. Gradient descent techniques

Suppose we have four training examples under the two-category case, i.e.

$$D = \{(x_i, \omega_i) \mid 1 \leqslant i \leqslant 4\} \tag{5}$$

where

- $x_1 = (1,1)^T, \quad \omega_1 = -1$
- $x_2 = (2,1)^T, \quad \omega_2 = -1$
- $x_3 = (-1,-2)^T, \quad \omega_3 = +1$
- $x_4 = (2,-3)^T, \quad \omega_4 = +1$

A **linear discriminant function**:

$$g(x) = w^T x + b \tag{6}$$

is adopted to learn from the training examples, and the **criterion function** to be minimized is:

$$J(w,b) = \frac{1}{2} w^T w - \sum_{i=1}^{4} \omega_i g(x_i) \tag{7}$$

Given the initial model:

- $w_0 = (2,1)^T$
- $b_0 = -1$

Answer the following questions:

(a) If **gradient descent techniques** are utilized to minimize the criterion function, what is the resulting discriminant function **after three gradient-descent steps** with learning rate $\eta = 0.1$ and threshold $\epsilon = 10^{-6}$? *(12 points)*

(b) Given a pattern $x = (-2,-2)^T$, should we decide **positive (+1)** or **negative (-1)** on x? *(3 points)*

(a.)

$$x_1 = (1,1)^T, \quad \omega_1 = -1$$
$$x_2 = (2,1)^T, \quad \omega_2 = -1$$
$$x_3 = (-1,-2)^T, \quad \omega_3 = +1$$
$$x_4 = (2,-3)^T, \quad \omega_4 = +1$$

$$J(w,b) = \frac{1}{2} w^T w - \sum \omega_i (w^T x_i + b)$$
$$= \frac{1}{2} w^T w - w^T \sum \omega_i x_i - b \sum \omega_i$$

$$\nabla_w J = w - \sum \omega_i x_i$$
$$\nabla_b J = -\sum \omega_i = 0$$

Since the gradient of b does not change,

$$b_0 = b_1 = b_2 = b_3 = -1$$

$$\sum \omega_i x_i = -1(1,1)^T - 1(2,1)^T + 1(-1,-2)^T + 1(2,-3)^T = (-2,-7)$$

Step 1

$$\nabla_w J_0 = w_0 - (-2,-7)^T$$
$$= (2,1)^T - (-2,-7)^T$$
$$= (4,8)$$

$$w_1 = w_0 - 0.1 \times \nabla_w J_0$$
$$= (2,1)^T - (0.4, 0.8)^T$$
$$= (1.6, 0.2)$$

Step 2

$$\nabla_w J_1 = w_1 - (-2,-7)^T$$
$$= (1.6, 0.2)^T - (-2,-7)^T$$
$$= (3.6, 7.2)$$

$$w_2 = w_1 - 0.1 \times \nabla_w J_1$$
$$= (1.6, 0.2)^T - (0.36, 0.72)^T$$
$$= (1.24, -0.52)$$

Step 3

$$\nabla_w J_2 = w_2 - (-2,-7)^T$$
$$= (1.24, -0.52)^T - (-2,-7)^T$$
$$= (3.24, 6.48)$$

$$w_3 = w_2 - 0.1 \times \nabla_w J_2$$
$$= (1.24, -0.52)^T - (0.324, 0.648)^T$$
$$= (0.916, -1.168)$$

$$\therefore \text{ Final discriminant function: } g(x) = 0.916\, x_1 - 1.168\, x_2 - 1$$

Q9. Consider a two-dimensional two-class classification problem, where

$$p(x|\omega_1) \sim N(0, I), \qquad p(x|\omega_2) \sim N\left(\begin{pmatrix}1\\1\end{pmatrix}, I\right),$$

and $P(\omega_1) = P(\omega_2) = 1/2$.

(a) Compute the Bayes decision boundary.

(b) For the same prior probabilities, but

$$p(x|\omega_1) \sim N\left(0, \begin{pmatrix}20 & 5\\5 & 2\end{pmatrix}\right), \qquad p(x|\omega_2) \sim N\left(\begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}5 & 4\\4 & 5\end{pmatrix}\right),$$

repeat the above steps.

(a) $P(\omega_1) = P(\omega_2) = 0.5$

Category 1:

$$p(x \mid \omega_1) \sim \mathcal{N}(\mu_1, \Sigma_1), \qquad \mu_1 = \begin{pmatrix}0\\0\end{pmatrix}, \qquad \Sigma_1 = I = \begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}.$$

Category 2:

$$p(x \mid \omega_2) \sim \mathcal{N}(\mu_2, \Sigma_2), \qquad \mu_2 = \begin{pmatrix}1\\1\end{pmatrix}, \qquad \Sigma_2 = I = \begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}.$$

Discriminant function: $g(x) = w_i^T x + w_{i0}$

where

$$w_i = \Sigma^{-1} \mu_i^T, \qquad w_{i0} = -\tfrac{1}{2} \mu_i^T \Sigma^{-1} \mu_i + \ln P(\omega_i)$$

Thus,

$$w_1 = I \begin{pmatrix}0\\0\end{pmatrix} = (0,0)$$

$$w_{10} = -\tfrac{1}{2}(0,0)\, I \begin{pmatrix}0\\0\end{pmatrix} + \ln \tfrac{1}{2} = \ln \tfrac{1}{2}$$

$$w_2 = I \begin{pmatrix}1\\1\end{pmatrix} = (1,1)$$

$$w_{20} = -\tfrac{1}{2}(1,1)\, I \begin{pmatrix}1\\1\end{pmatrix} + \ln \tfrac{1}{2} = -1 + \ln \tfrac{1}{2}$$

Decision boundary is determined by

$$g_1(x) = g_2(x).$$

$$x_1 + x_2 = 1.$$

Hence, the Bayes decision boundary is

$$x_1 + x_2 = 1.$$

(b) Σ_i is arbitrary.

The discriminant function is

$$g_i(x) = x^T W_i x + w_i^T x + w_{i0},$$

where

$$W_i = -\tfrac{1}{2} \Sigma_i^{-1}, \qquad w_i = \Sigma_i^{-1} \mu_i,$$
$$w_{i0} = -\tfrac{1}{2} \mu_i^T \Sigma_i^{-1} \mu_i - \tfrac{1}{2} \ln |\Sigma_i| + \ln P(\omega_i).$$

Given $P(\omega_1) = P(\omega_2) = 0.5$.

For class ω_1:

$$\mu_1 = \begin{pmatrix}0\\0\end{pmatrix}, \quad \Sigma_1 = \begin{pmatrix}2 & 0.5\\0.5 & 2\end{pmatrix}.$$

The determinant is

$$|\Sigma_1| = \begin{vmatrix}2 & 0.5\\0.5 & 2\end{vmatrix} = \frac{15}{4},$$

and

$$ad - bc = 4 - 0.25 = \frac{15}{4}.$$

Thus,

$$\Sigma_1^{-1} = \frac{4}{15}\begin{pmatrix}2 & -\frac{1}{2}\\-\frac{1}{2} & 2\end{pmatrix} = \begin{pmatrix}\frac{8}{15} & -\frac{2}{15}\\-\frac{2}{15} & \frac{8}{15}\end{pmatrix}.$$

Then

$$W_1 = -\tfrac{1}{2}\Sigma_1^{-1} = \begin{pmatrix}-\frac{4}{15} & \frac{1}{15}\\\frac{1}{15} & -\frac{4}{15}\end{pmatrix}, \qquad w_1 = \Sigma_1^{-1}\mu_1 = \begin{pmatrix}0\\0\end{pmatrix}.$$

$$w_{10} = -\tfrac{1}{2}\mu_1^T \Sigma_1^{-1} \mu_1 + \ln \tfrac{1}{2} - \tfrac{1}{2} \ln |\Sigma_1| = \ln \tfrac{1}{2} - 0.66.$$

Thus,

$$g_1(x) = x^T W_1 x + w_1^T x + w_{10}$$
$$= (x_1, x_2)\begin{pmatrix}-\frac{4}{15} & \frac{1}{15}\\\frac{1}{15} & -\frac{4}{15}\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix} + \ln \tfrac{1}{2} - 0.66,$$
$$= -\frac{4}{15}x_1^2 + \frac{2}{15}x_1 x_2 - \frac{4}{15}x_2^2 + \ln \tfrac{1}{2} - 0.66.$$

For class ω_2:

$$\mu_2 = \begin{pmatrix}1\\1\end{pmatrix}, \quad \Sigma_2 = \begin{pmatrix}5 & 4\\4 & 5\end{pmatrix}.$$

The determinant is

$$|\Sigma_2| = 25 - 16 = 9,$$

and hence

$$\Sigma_2^{-1} = \frac{1}{9}\begin{pmatrix}5 & -4\\-4 & 5\end{pmatrix} = \begin{pmatrix}\frac{5}{9} & -\frac{4}{9}\\-\frac{4}{9} & \frac{5}{9}\end{pmatrix}.$$

Then

$$W_2 = -\tfrac{1}{2}\begin{pmatrix}\frac{5}{9} & -\frac{4}{9}\\-\frac{4}{9} & \frac{5}{9}\end{pmatrix} = \begin{pmatrix}-\frac{5}{18} & \frac{2}{9}\\\frac{2}{9} & -\frac{5}{18}\end{pmatrix}.$$

$$w_2 = \Sigma_2^{-1}\mu_2 = \begin{pmatrix}\frac{1}{9}\\\frac{1}{9}\end{pmatrix}.$$

$$w_{20} = -\tfrac{1}{2}\mu_2^T \Sigma_2^{-1} \mu_2 + \ln \tfrac{1}{2} - \tfrac{1}{2} \ln 9$$
$$= -\tfrac{1}{2}\left(\tfrac{2}{9}\right) + \ln \tfrac{1}{2} - \ln 3.$$

Thus,

$$g_2(x) = x^T W_2 x + w_2^T x + w_{20}$$
$$= (x_1, x_2)\begin{pmatrix}-\frac{5}{18} & \frac{2}{9}\\\frac{2}{9} & -\frac{5}{18}\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix} + \frac{1}{9}(x_1 + x_2) + \ln \tfrac{1}{2} - \ln 3.$$

Setting the decision boundary $g_1(x) = g_2(x)$ yields

$$x_1^2 + x_2^2 - 28x_1x_2 - 10x_1 - 10x_2 + 50 = 0.$$